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Endoscopic tool for debriding and removing polyps

Abstract

An improved endoscopic tool easily and efficiently obtains samples of multiple polyps from a patient by debriding one or more polyps and retrieving the debrided polyps without having to alternate between using a separate cutting tool and a separate sample retrieving tool and may be used with an endoscope.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 17/234,270, filed Apr. 29, 2021, titled “ENDOSCOPIC TOOL FOR DEBRIDING AND REMOVING POLYPS,” which is a continuation application of U.S. application Ser. No. 15/804,884, titled “ENDOSCOPIC TOOL FOR DEBRIDING AND REMOVING POLYPS,” filed Nov. 6, 2017, which is a continuation application of U.S. application Ser. No. 13/336,491, titled, “ENDOSCOPIC TOOL FOR DEBRIDING AND REMOVING POLYPS,” filed Dec. 23, 2011, which application claims priority to U.S. Application No. 61/566,472, titled, “ENDOSCOPIC TOOL FOR DEBRIDING AND

REMOVING POLYPS,” filed Dec. 2, 2011, all of which are hereby incorporated by reference in their entireties.

BACKGROUND OF THE INVENTION

(1) Colon cancer is the third leading cause of cancer in the United States but is the second leading cause of cancer-related deaths. Colon cancer arises from pre-existing colon polyps (adenomas) that occur in as many as 35% of the US population. Colon polyps can either be benign, precancerous or cancerous. Colonoscopy is widely regarded as an excellent screening tool for colon cancer that is increasing in incidence worldwide. According to the literature, a 1% increase in colonoscopy screening results in a 3% decrease in the incidence of colon cancer. The current demand for colonoscopy exceeds the ability of the medical system to provide adequate screening. Despite the increase in colon cancer screening the past few decades, only 55% of the eligible population is screened, falling far short of the recommended 80%, leaving 30 million patients at risk.

(2) Due to the lack of adequate resources, operators performing an colonoscopy typically only sample the largest polyps, exposing the patient to sample bias by typically leaving behind smaller polyps that could advance to colon cancer prior to future colonoscopy. Because of the sample bias, a negative result from the sampled polyps does not ensure the patient is truly cancer-free.

Furthermore, operators are not incentivized to remove more polyps since the reimbursement is not based on the number of polyps removed or the adequacy of resection of any particular polyp.

Existing polyps removal techniques are cumbersome and time consuming, and not reimbursed at a higher level.

(3) At present, colon polyps are removed using a snare that is introduced into the patient's body via a working channel defined within an endoscope. The tip of the snare is passed around the stalk of the polyp to cut the polyp from the colon wall. Once the cut has been made, the cut polyp lies on the intestinal wall of the patient until it is retrieved by the operator as a sample. To retrieve the sample, the snare is first removed from the endoscope and a biopsy forceps is fed through the same channel of the endoscope to retrieve the sample. The limiting factor for patient access to colonoscopy is the amount of time required to complete procedures where patients have colon polyps.

(4) Accordingly, there is a need for an improved endoscopic tool that increases the precision and speed of polyp removal for biopsy.

SUMMARY OF THE INVENTION

(5) An improved endoscopic tool is provided that can easily and efficiently obtain samples of multiple polyps from a patient. In particular, the improved endoscopic tool is capable of debriiding one or more polyps and retrieving the debriided polyps without having to alternate between using a separate cutting tool and a separate sample retrieving tool. The sampling can be integrated with colonoscopy inspection.

(6) In one aspect, an endoscopic biopsy retrieval tool adapted for use with an endoscope a housing, a debriiding component coupled to the housing, and a sample retrieval conduit disposed within the housing for retrieving debriided material that is debriided by the debriiding component. In various embodiments, an improved endoscope may be configured with an integrated endoscopic biopsy retrieval tool that includes a debriiding component and a sample retrieval conduit for retrieving debriided material that is debriided by the debriiding component.

(7) In another aspect, a method of retrieving polyps from a patient's body includes disposing an endoscopic tool within an instrumentation channel of an endoscope, inserting the endoscope in a patient's body, actuating a debriiding component of the endoscopic tool to cut a polyp within the patient's body, and actuating a sample retrieval component of the endoscopic tool to remove the cut polyp from within the patient's body.

(8) In yet another aspect, An endoscope includes a first end and a second end separated by a flexible housing. An instrumentation channel extends from the first end to the second end and an

endoscopic tool is coupled to the instrumentation channel at the first end of the endoscope. The endoscopic tool includes a debriding component and a sample retrieval conduit partially disposed within the instrumentation channel.

(9) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended that this Summary be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that offer any or all advantages or solve any or all state of the art problems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is illustratively shown and described in reference to the accompanying drawing in which:

(2) FIG. 1 illustrates a perspective partial view of an endoscope according to embodiments of the present disclosure;

(3) FIGS. 2A and 2B illustrate side perspective views of an endoscopic tool coupled with the endoscope shown in FIG. 1 according to embodiments of the present disclosure;

(4) FIGS. 3A and 3B illustrate side perspective views of an alternate endoscopic tool coupled with the endoscope shown in FIG. 1 according to embodiments of the present disclosure;

(5) FIG. 4A illustrates an exploded view of the endoscopic tool that can be coupled with the endoscope according to embodiments of the present disclosure;

(6) FIG. 4B illustrates a perspective view diagram of the endoscopic tool coupled to the endoscope illustrating the various conduits associated with the endoscopic tool;

(7) FIG. 5 illustrates a side perspective view of another alternate endoscopic tool coupled with the endoscope shown in FIG. 1 according to embodiments of the present disclosure;

(8) FIG. 6 illustrates an enlarged view of an alternate endoscopic tool according to embodiments of the present disclosure;

(9) FIG. 7 illustrates a perspective view of an outer blade of a cutting tool of the endoscopic tool shown in FIG. 6 according to embodiments of the present disclosure;

(10) FIG. 8 illustrates a perspective view of an inner blade of the cutting tool of the endoscopic tool shown in FIG. 6 according to embodiments of the present disclosure;

(11) FIG. 9 illustrates a perspective view of a rotor of the endoscopic tool shown in FIG. 6 according to embodiments of the present disclosure;

(12) FIG. 10 illustrates a perspective view of a casing of the endoscopic tool shown in FIG. 6 according to embodiments of the present disclosure;

(13) FIG. 11 illustrates a perspective view of a cap of the endoscopic tool shown in FIG. 6 according to embodiments of the present disclosure;

(14) FIG. 12 illustrates a perspective view of a coupling member of the endoscopic tool shown in FIG. 6 according to embodiments of the present disclosure;

(15) FIG. 13 illustrates a perspective view diagram of the endoscopic tool coupled to the endoscope illustrating the various conduits associated with the endoscopic tool;

(16) FIG. 14 illustrates another perspective view diagram of the endoscopic tool coupled to the endoscope illustrating the various conduits associated with the endoscopic tool; and

(17) FIG. 15 is a conceptual system architecture diagram illustrating various components for operating the endoscopic tool according to embodiments of the present disclosure.

DETAILED DESCRIPTION

(18) Technologies provided herein are directed towards an improved endoscopic tool that can easily

and efficiently obtain samples of multiple polyps from a patient. In particular, the improved endoscopic tool is capable of debriding samples from one or more polyps and retrieving the debrided samples without having to remove the endoscopic tool from the treatment site within the patient's body.

(19) Presently, physicians typically use a snare device that is capable of cutting polyps grown within the patient's body. The snare device, however, is not capable of retrieving the cut polyps. As such, in order for a physician to remove polyps from within a patient's body, the physician has to insert the snare device through the working channel of an endoscope, cut the polyps desired to be cut by the physician and leave the cut polyp at or around the surgical site within the patient's body and then remove the snare device from the working channel. Once the snare device is removed, the physician then inserts a sample retrieving device, such as forceps, and removes the cut polyps from the surgical site by grasping the cut polyp and removing the forceps from the working channel of the endoscope and releasing the cut polyp outside the patient's body. To avoid contaminating the cut polyps, the physician may remove cut polyps one at a time. As one can imagine, this is a very inefficient and time consuming process, while at the same time, may not allow the physician to identify a particular location within the patient's body a particular retrieved polyp had grown. As a result, if any of the retrieved polyps is determined to be cancerous, the physician may not be able to determine the location from which the cancerous polyp was removed hence complicating treatment planning.

(20) The present disclosure will be more completely understood through the following description, which should be read in conjunction with the drawings. In this description, like numbers refer to similar elements within various embodiments of the present disclosure. Within this description, the claims will be explained with respect to embodiments. The skilled artisan will readily appreciate that the methods, apparatus and systems described herein are merely exemplary and that variations can be made without departing from the spirit and scope of the disclosure.

(21) Referring now to the drawings, FIG. 1 illustrates a perspective partial view of an endoscope according to embodiments of the present disclosure. Although the present disclosure is directed towards endoscopic tools adapted for use with any type of endoscope, for sake of convenience, the teachings of the present disclosure are directed towards endoscopic tools used with a lower GI scope, such as a colonoscope. It should, however, be appreciated that the scope of the present disclosure is not limited to endoscopic tools for use with GI scopes, but extends to any type of endoscope, including but not limited to gastroscopes and laryngoscopes, or other medical devices that may be used to treat patients.

(22) According to various embodiments, a typical lower GI scope **100** includes a flexible body portion that extends from a first end or head portion **102** to a second end or handle portion. The head portion **102** may be configured to swivel so as to orient a tip **104** of the head portion **102** in any direction within a hemispherical space. The handle portion (not shown) has controls that allows the operator of the endoscope **100** to steer the colonoscope towards an area of interest within the colon and turn the corners between colon segments with two steering wheels.

(23) A series of instruments reside on the face **106** of the scope's tip **104**, including but not limited to, one or more water channels **108A-N**, generally referred to as water channels **108**, for irrigating the area with water, one or more light sources **110A-N**, generally referred to as light sources **110**, a camera lens **112**, and an instrument channel **120** through which an endoscopic tool can be passed through to conduct a number of operations. The instrumentation channel **120** can vary in size based on the type of endoscope **100** being used. In various embodiments, the diameter of the instrumentation channel **120** can range from 3.2 mm to 4 mm. Some larger scopes may have two instrumentation channels **120** so that two tools can be passed into the patient simultaneously. However, larger scopes may cause discomfort to the patient and may be too large to enter the patient's body through some of the smaller cavities.

(24) FIGS. 2A and 2B and 3A and 3B illustrate side perspective views of an endoscopic tool

coupled with the endoscope shown in FIG. 1 according to embodiments of the present disclosure. The endoscopic tool **220** is configured to be fed through the instrumentation channel **120** of the endoscope **100**. As shown in FIGS. 2A and 2B, the endoscopic tool **220** is capable of extending outside the tip **104** of the endoscope **100**, while FIGS. 3A and 3B show that the endoscopic tool **220** can be retracted within the endoscope such that no part of the endoscopic tool **220** is extending beyond the tip **104** of the endoscope **100**. As will be described in further detail with respect to FIG. 4, the endoscopic tool **220** is capable of cutting or debriding a polyp as well as obtaining the debrided polyp from the surgical site without having to remove the endoscopic tool **220** from the endoscope **100**.

(25) FIG. 4A illustrates an exploded view of the endoscopic tool **220** adapted for use with the endoscope **100** according to embodiments of the present disclosure. The endoscopic tool **220** includes a debriding component for debriding polyps grown in the patient's body, and a sample retrieval component for retrieving the debrided polyps from the surgical site. The endoscopic tool includes a tubing **410** coupled to a cap **420**. In various embodiments, the cap **420** may be sealingly engaged with the tubing **410**. The cap is coupled to a spindle **430** at a first portion of the spindle **430**. In various embodiments, the spindle **430** may be at least substantially hollow. The spindle **430** is coupled to a rotor **440**, which is configured to rotate the spindle **430**. A second portion of the spindle **430** includes an inner blade **450** that may be configured to interact with an outer blade **460**. The outer blade **460** is separated from the inner blade by an irrigation channel (not shown). A casing **470** is configured to encompass the cap **420** and the rotor **440**, as shown above with respect to FIGS. 2A and 3A. It should be appreciated that other components, such as washers, bearings, seals, and the like, may be included in the endoscopic tool **220**.

(26) FIG. 4B is a schematic diagram of a full endoscope including the endoscopic tool or tube tool for insertion into an inspection area via body cavity investigation or endoscopic surgery channel showing good combination of viewing, lighting, the description of distal elements by proximal end adjustment, etc. In various embodiments, the cap, connector, rotor and casing may be made from injection molded plastic. The spindle and the cannula may be made from surgical grade steel, and the tubing may be made from silicone. However, it should be appreciated that these materials are merely examples of materials that can be used. Those skilled in the art will appreciate that alternate materials may be used instead of the ones described above.

(27) The tubing **410** in FIG. 4A may be sized to pass through the instrumentation channel **120** of the endoscope **100** in FIGS. 4A and 4B. The tubing **410** may include one or more pneumatic fluid entry conduits **412**, one or more pneumatic fluid exit conduits **414**, one or more irrigation conduits **416**, and one or more suction conduits **418**. The pneumatic fluid entry conduits **412** are configured to supply pressurized air to pneumatically drive the rotor **440**, while the pneumatic fluid exit conduits **414** remove the air supplied by the pneumatic fluid entry conduits **412** to prevent a large amount of air from entering the patient's body. The irrigation conduits **416** supply an irrigation fluid, such as water, between the inner blade **450** and the outer blade **460** to help lubricate the area between the inner blade **450** and the outer blade **460**. In addition, the irrigation fluid then flows from the outside of the inner blade **450** to the inside portion of the inner blade **450**. It should be appreciated that the inside portion of the inner blade **450** may be coupled to the suction conduit **418** of the tubing **410** via the cap **420**. The irrigation fluid that flows through the inside portion of the inner blade **450** and the suction conduit **418** helps lubricate the suction conduit **418**, through which the debrided polyps and other waste from the patient's body are removed. As described above, the tubing **410** is coupled to the cap **420** at a first end, but is coupled to one or more components at a second end (not shown). For instance, at the second end, the pneumatic air entry conduits **412** may be coupled to a compressed air source, while the irrigation fluid conduit **416** may be coupled to a water supply source. In addition, the pneumatic fluid exit conduits **414** may be coupled to the compressed air source or simply left exposed outside the patient's body for venting.

(28) In various embodiments, the suction conduit **418** may be coupled to a disposable cartridge that

is configured to catch the cut polyps and store them for examination at a later time. In various embodiments, the disposable cartridge may include multiple collection bins. The operator may be capable of selecting the collection bin in which to collect a sample of a particular cut polyp. Upon selecting the collection bin, the suction conduit **418** supplies the collected material from within the patient's body to the particular collection bin. As such, the operator may be able to collect samples for each polyp in individual collection bins. In this way, the cancerous nature of individual polyps can be determined.

(29) The cap **420** may be sized to fit within the first end of the tubing **410**. In various embodiments, the first end of the tubing **410** may include a connector that is configured to couple with the cap **420**. In various embodiments, the cap **420** may be press fitted into the connector of the tubing **410**. As such, the cap **420** may include corresponding conduits that match the conduits of the tubing **410**. Accordingly, compressed air from the compressed air source may be supplied through the pneumatic air entry conduits **412** of the tubing **410** and corresponding pneumatic air entry conduits of the cap **420** towards the rotor **440**. The rotor **440** may include one or more rotor blades **442** on which the compressed air is impinged thereby causing the rotor **440** to rotate. The air impinging on the rotor blades **442** may then exit through the corresponding pneumatic air exit conduits of the cap and the pneumatic air entry conduits **414** of the tubing **410**. The speed at which the rotor **440** can rotate depends on the amount of air and the pressure at which the air is supplied to the rotor **440**. In various embodiments, the speed at which the rotor **440** rotates may be controlled by the operator of the endoscope **100**. Although the present disclosure discloses pneumatic means for operating the rotor, alternate embodiments may include hydraulic means for operating the rotor. In such embodiments, a fluid, such as water, may be supplied in lieu of compressed air, in the pneumatic air entry conduit **412**.

(30) As described above, the spindle **430** is coupled to the rotor **440**, such that when the rotor **440** rotates, the spindle **430** also rotates. In various embodiments, the first end of the spindle **430** includes the inner blade **450**, which correspondingly, also rotates along with the rotor **440**. The inner blade **450** may be sized to fit within the diameter of the outer blade **460**. In various embodiments, irrigation fluid supplied from an irrigation fluid source may be supplied through the irrigation fluid conduit **416** of the tubing **410** and the corresponding conduit of the cap **420**, along the space between the inner blade **450** and the outer blade **460**, and into the suction conduit **418** defined by the inner diameter of the inner blade **450**. It should be appreciated that since the suction conduit **418** is coupled to a vacuum source, fluids and other material may be suctioned through the suction conduit. In this way, the irrigation fluid is able to lubricate at least a substantial length of the suction conduit **418**, from the tip **452** of the inner blade **450**, through the spindle **430**, cap **420**, and tubing **410** into the disposable cartridge described above.

(31) The inner blade **450** may rotate relative to the outer blade **460** such that the interaction between the inner blade **450** and the outer blade **460** causes polyps to be cut upon contact with the inner blade **450**. In various embodiments, other mechanisms for cutting polyps may be utilized, which may or may not include the use of a rotor **440**, inner blade **450** or outer blade **460**.

(32) The debriding component may generally be configured to debride a polyp. The term debride may be defined herein to refer to any action involving detaching the polyp from a surface of the patient's body. Accordingly, actions, including but not limited to, cutting, snaring, shredding, slicing, shattering, either entirely or partially, are also considered to lie within the definition of the term debride. Accordingly, the debriding component may be a component that is capable of cutting, snaring, shredding, slicing, shattering, a polyp from a surface of the patient's body. As such, the debriding component may be implemented as a forceps, scissor, knife, snare, shredder, or any other component that can debride a polyp. In some embodiments, the debriding component may be manually actuated such that the debriding component may be operated through the translation of mechanical forces exerted by an operator or automatically actuated, using a turbine, electrical motor, or any other force generating component to actuate the debriding component. For instance,

the debridging component may be actuated hydraulically, pneumatically, or electrically. In various embodiments, a separate conduit passing through the tubing or a channel of the endoscope may be configured to carry an electrical wire to provide power to the electrically powered actuator, such as an electrical motor.

(33) According to various embodiments, the debridging component may include a turbine assembly, which is made up of the rotor **440**, the rotor blades **442**, and the spindle **430**. The operator may actuate the debridging component of the endoscopic tool by supplying compressed air to the turbine assembly. When the operator is ready to begin debridging the polyp, the operator actuates the turbine assembly causing the debridging component to be actuated. In embodiments, such as the embodiment disclosed in FIG. 4, actuating the debridging component may constitute causing the inner blade **450** to rotate relative to the outer blade **460**. Upon actuation, the operator may bring the endoscopic tool **220** towards the polyp to be debrided causing the inner blade **450** to debride the polyp, causing portions of the debrided polyp to lie in the vicinity around the area where the polyp had grown. The operator may then de-actuate the turbine assembly and actuate suction through the suction conduit **418**. The operator may then bring the inner blade close to the cut polyp causing the cut polyp to be retrieved through the suction conduit **418**. In various embodiments, the suction component of the endoscopic tool may be actuated while the debridging component is actuated, thereby allowing any debrided material to be retrieved by the suction component.

(34) Although the above embodiment houses a debridging component that utilizes a turbine assembly, the scope of the present disclosure is not limited to such embodiments. Rather, it should be appreciated by those skilled in the art that the debridging component may be manually operated or may utilize any other means of debridging a polyp such that the debrided polyps are capable of being retrieved from the surgical site via the suction conduit described above. Accordingly, examples of debridging components may include, but are not limited to, snips, blades, saws, or any other sharp tools that may or may not be driven by a turbine assembly. It should be appreciated that using a debridging component that is able to cut a polyp into small enough pieces may be desirable such that the cut pieces may be retrieved via the suction conduit without having to remove the endoscopic tool from the endoscope.

(35) The geometry and assembly of the turbine assembly for rotating at least one of the cutting tool blades may be based on fluid dynamics. Bernoulli's equation can be used to explain the conversion between fluid pressure and the fluid velocity. According to this equation, the fluid velocity is related to the initial fluid pressure by the equation:

$$(36) V = \sqrt{2 * \frac{P}{D}}$$

where V is Velocity, P is Pressure, and D is Mass density.

(37) In order for the fluid to reach the calculated velocity, the fluid can be developed at the point of exit such that the channel through which the fluid is flowing meets an empirically determined L/D ratio of 2, where 'D' is the wetted diameter of the flow and the 'L' is the length of the channel.

(38) To further understand the interaction of the rotor blades and the fluid, it is assumed that the rotor blade is made so that the air jet impinges the rotor blade on a plane. The equation of linear momentum can be applied to find the forces generated:

$$(39) \text{.Math. } F = \frac{d}{dt}(\int \int \int V_p * dVol.) + \text{.Math. } (\dot{m}V)_{out} - \text{.Math. } (\dot{m}V)_{in}$$

where: m is the mass flow of the impinging air jet, and V is Volume.

(40) Assuming that the control volume remains constant (volume between blades), the force created on the blade can be solved for:

$$(41) \text{.Math. } F = \dot{m}(V_{out} - V_{in})$$

(42) The quantity V.sub.out and V.sub.in are the same in an impulse turbine, the momentum change being created by the changing direction of the fluid only. The mass flow {dot over (m)} is defined by the pump that is to be specified. The actual numerical value also needs to account for the velocity of the rotor. So finally, the force generated by a single blade-air jet interaction is:

$$(43) \text{ .Math. } F = \dot{m}(V_{\text{jet}} - V_{\text{rotor}}) - (V_{\text{jet}} - V_{\text{rotor}})\cos(\theta) \text{ .Math. } F = \dot{m}(V_{\text{jet}} - V_{\text{rotor}})(1 - \cos(\theta))$$

where 'θ' is the difference of the angle between the incoming air jet to that of the exiting air jet. Thought theoretically, the maximum amount of torque can be generated by a 'θ' value of 180°, but doing so will actually send the incoming jet onto the back of the following blade. Accordingly, the angle is best given a design value 15° to 20° below 180 to allow a fluid a clean exit. Finally, the force can be defined into a rotational torque:

$$(44) \text{ .Math. } T = (\dot{m} / r)(V_{\text{jet}} - V_{\text{rotor}})(1 - \cos(\theta))$$

(45) A second force that must be considered comes from redirecting the air jet from the nozzle into the turbine wheel. To power the turbine, the air jet must be turned 90° into the direction of the blades from the direction of the air jet. The turning of the air jet will create a force on the stationary housing that is a function of the jet velocity, which in turn is proportional to the applied pressure:

$$(46) \text{ .Math. } F = \dot{m}V_{\text{jet}}$$

(47) This force must be reacted by the connection between the housing and the endoscope, a failure to do so can result in the ejection of the turbine assembly during operation.

(48) Computational analyses based on Finite Element Methods (FEM) reveal that the areas where the greatest stresses are found are located near the root of the blade where a sharp corner is located. The design of air input channel can be simplified by the existing air nozzle channel in endoscope. The air nozzle in existing endoscopes directs pressurized air across objective lens to remove moisture and also provides distension of a cavity being examined or directs pressurized water across objective lens to clear debris.

(49) Referring now to FIG. 4B, a perspective view diagram of the endoscopic tool coupled to the endoscope illustrating the various conduits associated with the endoscopic tool is shown. In particular, the pneumatic air entry conduit **412** is shown supplying pressurized air to the rotor assembly, while the pneumatic air exit conduit **412** (not shown in this view) removes the air from the rotor assembly to outside the endoscope **100**. The irrigation channel **416** is shown to carry irrigation fluid into the endoscopic tool **220**, where the irrigation fluid enters into the suction conduit **418**, which carries material from within the patient's body to a collection component outside the endoscope. As shown in FIG. 4B, the irrigation fluid may enter the suction conduit **418** at an irrigation fluid entry opening **419**. It should be appreciated that the placement of the irrigation fluid entry opening **419** may be placed anywhere along the suction conduit. Due to the suction force being applied to the suction conduit, irrigation fluid may be forced into the suction conduit without the risk of the materials flowing in the suction conduit from flowing outside the suction conduit through the irrigation fluid entry opening **419**. Moreover, in some embodiments, the irrigation channel may only supply irrigation fluid to the endoscopic tool while suction is being applied to the suction conduit.

(50) FIG. 5 illustrates a side perspective view of another alternate endoscopic tool coupled with the endoscope shown in FIG. 1 according to embodiments of the present disclosure. The add-on endoscopic tool **500** is sized to couple with the walls defining the instrumentation channel **120** of the tip **104** of the endoscope **100**. In various embodiments, the add-on endoscopic tool **500** may be removably attached to the instrumentation channel **120** of the endoscope **100** at the tip **104** of the endoscope **104** by way of an interference fit or a press fit. In other embodiments, the add-on endoscopic tool **500** may be coupled to the endoscope **100** using other attachment means known to those skilled in the art.

(51) Referring now to FIG. 6, an enlarged view of the add-on endoscopic tool **500** is shown. The add-on endoscopic tool includes an outer blade or support member **510**, an inner blade **520** disposed within the outer blade **510**, a rotor **530** coupled to the inner blade **520** and encompassed by a casing **540**. The casing is coupled to a cap **550**, which is further coupled to a connector **560**. In some embodiments, the connector **560** may be sized to engage with the inner diameter of the instrumentation channel **120** of the endoscope **100**. In alternate embodiments, any other component

of the endoscopic tool may be configured to engage with the endoscope **100** in such a manner as to secure the endoscopic tool to the instrumentation channel **120**.

(52) FIGS. **7-12** illustrate perspective views of the individual components of the add-on endoscopic tool shown in FIG. **6** according to embodiments of the present disclosure. In contrast to the endoscopic tool **220** disclosed with respect to FIGS. **1-4**, the add-on endoscopic tool **500** may be adapted to fit within a first end of instrumentation channel **120** of the endoscope **100**.

(53) In various embodiments, a second end of the instrumentation channel **120** may be coupled to a vacuum source, which causes material to be suctioned through the instrumentation channel **120**. A suction conduit extends from the vacuum source through the instrumentation channel of the endoscope, and further through the connector **560**, the cap **550**, and the rotor **530**, to a first end of the inner blade **520**, which has an opening defined by the inner diameter of the inner blade **520**. It should be appreciated that the connector **560**, the cap **550**, the casing **540**, and the rotor **530** have respective center bores **566**, **556**, **546** and **536** that are aligned such that materials are allowed to flow from the opening of the inner blade **520** to the vacuum source via the second end of the instrumentation channel **120**.

(54) In addition, the casing **540** of the add-on endoscopic tool **500** includes a pneumatic air entry port **542** and a pneumatic air exit port **544** as shown in FIG. **10**. The pneumatic air entry port **542** may be adapted to receive compressed air from a compressed air source through a pneumatic air entry conduit that passes along the length of the endoscope **100** to outside the patient's body, while the pneumatic air exit port **544** may be adapted to vent air that is impinged on the rotor **530** through a pneumatic air exit conduit that passes along the length of the endoscope **100** to outside the patient's body. In this way, the rotor may be actuated by supplying compressed air from the compressed air source, as described above with respect to FIGS. **1-4**. It should be appreciated that although the rotor and associated components disclosed herein describe the use of pneumatic air, the rotor may be driven hydraulically. In such embodiments, the pneumatic air conduits may be configured to carry a liquid, such as water, to and from the area around the rotor.

(55) Referring now also to FIG. **13**, it should be appreciated that the pneumatic air entry and exit conduits may extend from the add-on endoscopic tool to a pneumatic air source through the instrumentation channel **120** of the endoscope **100**. In such embodiments, a tubing that includes separate conduits for the pneumatic air entry and exit conduits and the suction conduit may extend from outside the endoscope to the add-on endoscopic tool within the endoscope. The tubing may be capable of being fed through the instrumentation channel of the endoscope and coupled to the add-on endoscopic tool **500**. In such embodiments, the add-on endoscopic tool **500** may be configured with an additional component that has predefined channels that couple the respective channels of the tubing with the associated with the pneumatic air entry and exit openings of the add-on endoscopic tool and the suction conduit formed within the add-on endoscopic tool. In addition, an irrigation fluid channel may also be defined within the tubing such that irrigation fluid may be supplied to the add-on endoscopic tool **500**, from where the irrigation fluid is diverted into the suction conduit.

(56) In various embodiments, the tip of the outer blade **510** may be sharp and may cause discomfort to the patient while entering a cavity of the patient's body. As such, a guard structure (not shown), such as a gel cap or other similar structure, may be attached to the outer blade prior to inserting the add-on endoscopic tool into the patient's body to prevent injuries from the outer blade contacting a surface of the patient's body. Once the endoscopic tool is inserted in the patient's body, the guard structure may be released from the outer blade **510**. In various embodiments, the guard structure may dissolve upon entering the patient's body.

(57) Referring now to FIG. **14**, an improved endoscope having a built in polyp removal assembly is shown according to embodiments of the present disclosure. The improved endoscope **1400** may be similar to conventional endoscopes in many aspects, but may differ in that the improved endoscope may include a built in polyp removal assembly **1440** within an instrumentation channel of the

endoscope **1400**. The polyp removal assembly **1440** may include a turbine assembly having a rotor **1442** with rotor blades sealed in a casing **1444** that has one or more inlet and outlet ports for allowing either pneumatic or hydraulic fluid to actuate the rotor **1442**. The inlet ports may be designed such that the fluid may interact with the rotor blades at a suitable angle to ensure that the rotor can be driven at desired speeds.

(58) In addition, the polyp removal assembly **1440** may be coupled to a connector **1420**, which is configured to couple the polyp removal assembly **1440** to a tubing **1470**. The tubing **1470** may include a pneumatic air entry conduit **1412**, a pneumatic air exit conduit (not shown), an irrigation fluid conduit **1416** and a suction conduit **1418** that passes through the center of the turbine assembly. The tubing **1440** may be sized such that the tubing **1440** can be securely coupled to the connector **1420** such that one or more of the conduits of the tubing **1440** are coupled to corresponding conduits within the connector **1440**. The connector **1420** may be designed to include an irrigation fluid entry opening **419**, which allows irrigation fluid to pass into the suction conduit **1418** of the tubing **1440** when the tubing is coupled to the connector.

(59) The turbine assembly of the endoscope **1400** may be configured to couple with a removable debriding assembly **1460**, which includes a spindle and a cannula, in a manner that causes the debriding assembly to be operational when the turbine assembly is operating.

(60) In other embodiments of the present disclosure, an endoscope may be designed to facilitate debriding one or more polyps and removing the debrided material associated with the polyps in a single operation. In various embodiments, the endoscope may include one or more separate channels for removing debrided material, supplying irrigation fluid, and supplying and removing at least one of pneumatic or hydraulic fluids. In addition, the endoscope may include a debriding component that may be fixedly or removably coupled to one end of the endoscope. In various embodiments, based on the operation of the debriding component, a separate debriding component channel may also be designed for the debriding component. In addition, the endoscope may include a light and a camera. In one embodiment, the endoscope may utilize existing channels to supply pneumatic or hydraulic fluids to the actuator of the endoscopic tool for actuating the debriding component. For instance, in the endoscope shown in FIG. **1**, the water channels **108A-N** may be modified to supply fluids to the actuator pneumatically or hydraulically. In such embodiments, the endoscopic tool may include a connector having a first end capable of being coupled to an opening associated with existing channels **108** of the endoscope, while another end of the connector is exposed to an opening at the actuator.

(61) In various embodiments of the present disclosure, the endoscopic tool may further be configured to detect the presence of tissue or muscle. This may be useful for physicians to take extra precautions to prevent bowel perforations while debriding polyps. In some embodiments, the endoscopic tool may be equipped with an electrical sensor that can communicate with a sensor processing component outside the endoscope to determine if a particular region of the patient's body is made from tissue or muscle. The sensor may gather temperature information as well as density information and provide signals corresponding to such to the sensor processing unit, which can determine from the signals, if the particular region is made from tissue or muscle.

(62) In addition, the endoscopic tool may be equipped with an injectable dye component through which a physician may mark a particular region within the patient's body. In other embodiments, the physician may mark a particular region utilizing the debriding component, without the use of an injectable dye.

(63) Although the present disclosure discloses various embodiments of an endoscopic tool, including but not limited to a tool that may be attached to the tip of the endoscope, and a tool that may be fed through the length of the endoscope, the scope of the present disclosure is not intended to be limited to such embodiments or to endoscopic tools in general. Rather, the scope of the present disclosure extends to any device that may debride and remove polyps from within a patient's body using a single tool. As such, the scope of the present disclosure extends to improved

endoscopes that may be built with some or all of the components of the endoscopic tools described herein. For instance, an improved endoscope with a built in turbine assembly and configured to be coupled to a debriding component is also disclosed herein. Furthermore, the endoscope may also include predefined conduits that extend through the length of the endoscope such that only the suction conduit may be defined by a disposable tubing, while the air entry and exit conduits and the irrigation conduit are permanently defined within the improved endoscope. In other embodiments, the suction conduit is also predefined but made such that the suction conduit may be cleaned and purified for use with multiple patients. Similarly, the debriding component may also be a part of the endoscope, but also capable of being cleaned and purified for use with multiple patients.

Furthermore, it should be understood by those skilled in the art that any or all of the components that constitute the endoscopic tool may be built into an existing endoscope or into a newly designed endoscope for use in debriding and removing polyps from within the patients body.

(64) Referring now to FIG. 15, a conceptual system architecture diagram illustrating various components for operating the endoscopic tool according to embodiments of the present disclosure is shown. The endoscopic system **1500** includes an endoscope **100** fitted with an endoscopic tool **220**, and which may be coupled to an air supply measurement system **1510**, an irrigation system **1530** and a polyp removal system **1540**. As described above, the tubing that extends within the endoscope **100** may include one or more pneumatic air entry conduits **412** and one or more pneumatic air exit conduits **414**. The pneumatic air entry conduits **412** are coupled to the air supply measurement system **1510**, which includes one or more sensors, gauges, valves, and other components to control the amount of gas, such as air, being supplied to the endoscope **100** to drive the rotor **440**. In some embodiments, the amount of air being supplied to the rotor **440** may be controlled using the air supply measurement system **1510**. Furthermore, delivery of the air to actuate the rotor **440** may be manually controlled by the physician using the endoscope **100**. In one embodiment, the physician may use a foot pedal or a hand-actuated lever to supply air to the rotor **440**.

(65) The pneumatic air exit conduit **414**, however, may not be coupled to any component. As a result, air exiting from the rotor **440** may simply exit the endoscope via the pneumatic air exit conduit **414** into the atmosphere. In alternate embodiments, the pneumatic air exit conduit **414** may be coupled to the air supply measurement system **1510** such that the air exiting the pneumatic air exit conduit **414** is supplied back to the rotor via the pneumatic air entry conduit **412**. It should be appreciated that a similar setup may be used for a hydraulically driven turbine system.

(66) The endoscope **100** may also be coupled to the irrigation system **1530** via the irrigation fluid conduit **416**. The irrigation system **1530** may include a flow meter **1534** coupled to an irrigation source **1532** for controlling the amount of fluid flowing from the irrigation source **1532** to the endoscope **100**.

(67) As described above, the endoscope **100** may also include a suction conduit **418** for removing polyps from within the patient's body. The suction conduit **418** may be coupled to the polyp removal system **1540**, which may be configured to store the polyps. In various embodiments, the physician may be able to collect samples in one or more cartridges **1542** within the polyp removal system **1540** such that the removed polyps can be tested individually.

(68) In various embodiments of the present disclosure, an endoscope, comprises a first end and a second end separated by a flexible housing, an instrumentation channel extending from the first end to the second end, and an endoscopic tool comprising a debriding component and a sample retrieval conduit disposed within the instrumentation channel. The endoscopic tool may further include a flexible tubing in which the sample retrieval conduit is partially disposed, the flexible tubing extending from the first end to the second end of the endoscope. The flexible tubing may also include a pneumatic air entry conduit and a fluid irrigation conduit. In various embodiments, the debriding component may include a turbine assembly and a cutting tool. In various embodiments in which the endoscope is configured to have a built in endoscopic tool, the instrumentation channel

may have a diameter that is larger than the instrumentation channels of existing endoscopes. In this way, larger portions of debrided material may be suctioned from within the patient's body without clogging the suction conduit.

(69) In other embodiments, an endoscope may include a first end and a second end separated by a flexible housing; an instrumentation channel extending from the first end to the second end; and an endoscopic tool coupled to the instrumentation channel at the first end of the endoscope, the endoscopic tool comprising a debriding component and a sample retrieval conduit partially disposed within the instrumentation channel. In some embodiments, the endoscopic tool may be removably attached to the endoscopic tool.

(70) In other embodiments of the present disclosure, an endoscopic system, includes an endoscope comprising a first end and a second end separated by a flexible housing and an instrumentation channel extending from the first end to the second end and an endoscopic tool coupled to the instrumentation channel at the first end of the endoscope. The endoscopic tool may include a debriding component and a flexible tubing having a length that is greater than the length of the endoscope. Moreover, the flexible tubing may include a sample retrieval conduit, an pneumatic air entry conduit, and a fluid irrigation conduit, a disposable cartridge configured to couple with the sample retrieval conduit proximal the second end of the endoscope, a pressurized air source configured to couple with the pneumatic air entry conduit proximal the second end of the endoscope, and a fluid irrigation source configured to couple with the fluid irrigation conduit proximal the second end of the endoscope. In various embodiments, the endoscope may also include at least one camera source and at least one light source. In some embodiments of the present disclosure, the pneumatic air entry conduit supplies pressurized air to a turbine assembly of the debriding component proximal the first end of the endoscope and the fluid irrigation conduit supplies irrigation fluid to the sample retrieval conduit proximal the first end of the endoscope.

(71) The present disclosure is illustratively described above in reference to the disclosed embodiments. Various modifications and changes may be made to the disclosed embodiments by persons skilled in the art without departing from the scope of the present disclosure as defined in the appended claims.

Claims

1. An endoscopic instrument sized to be inserted within an instrument channel of an endoscope, the endoscopic instrument comprising: an instrument head comprising a first proximal end and a first distal end configured to cut tissue within a subject; an actuator coupled with the first proximal end of the instrument head to drive the instrument head; and a flexible tubing coupled with the actuator such that the actuator is positioned between the instrument head and the flexible tubing, the flexible tubing and the actuator sized to be positioned within a distal portion of the instrument channel of the endoscope during use while the distal portion of the instrument channel is within a cavity of the subject.

2. The endoscopic instrument of claim 1, further comprising an engagement assembly, the engagement assembly configured to contact one or more walls of the instrument channel of the endoscope when actuated.

3. The endoscopic instrument of claim 2, wherein the engagement assembly includes a compliant ring structure configured to be deformed.

4. The endoscopic instrument of claim 1, wherein the actuator includes at least one of an electric motor, a tesla rotor, and a vane rotor.

5. The endoscopic instrument of claim 1, further comprising an energy storage component configured to power the actuator.

6. The endoscopic instrument of claim 1, wherein the endoscopic instrument has an outer diameter that is less than about 5 mm.

7. The endoscopic instrument of claim 1, wherein the actuator is one of a hydraulically powered actuator or a pneumatically powered actuator, and wherein the flexible tubing includes a fluid inlet tubular member configured to supply fluid to actuate the actuator and a fluid outlet tubular member configured to remove the fluid being supplied to actuate the actuator.
 8. The endoscopic instrument of claim 1, wherein the endoscopic instrument has an outer diameter that is 5 mm.
 9. The endoscopic instrument of claim 1, wherein the actuator is one of a hydraulically powered actuator, a pneumatically powered actuator or an electrically powered actuator.
 10. The endoscopic instrument of claim 1, wherein the instrument head includes an outer structure and a cutting shaft of the first distal end disposed within the outer structure, the cutting shaft coupled to the actuator and configured to rotate relative to the outer structure responsive to actuation of the actuator.
 11. The endoscopic instrument of claim 10, wherein the cutting shaft includes a hollow tubing and a material entry port configured to receive the cut tissue.
 12. The endoscopic instrument of claim 1, wherein the flexible tubing is at least 40 times as long as the instrument head.
 13. An endoscopic instrument, comprising: a cutting tool having a first distal end and a first proximal end, the first distal end of the cutting tool defining a material entry port configured to receive tissue; a turbine to drive the cutting tool, the turbine disposed within a portion of an instrument channel of an endoscope that is inserted within a cavity of a subject; a flexible tubing having a second proximal end defining a material exit port, the flexible tubing including: a fluid entry conduit configured to supply fluid to the turbine, and a fluid exit conduit configured to remove the fluid from the turbine.
 14. The endoscopic tool of claim 13, wherein the cutting tool comprises a cylindrical or part-cylindrical inner blade disposed within an outer blade, and an outer structure and a cutting shaft disposed within the outer structure, the cutting shaft coupled to the cutting tool and configured to rotate relative to the outer structure when the cutting tool is driven.
 15. The endoscopic tool of claim 14, wherein the turbine comprises a fluid driven rotor or vacuum driven rotor.
 16. The endoscopic tool of claim 15, wherein the fluid driven rotor or the vacuum driven rotor is coupled to the inner blade.
 17. The endoscopic tool of claim 15, wherein the turbine further comprises a hydraulic fluid exit conduit for removing irrigation fluid from the turbine and an irrigation fluid conduit for supplying the irrigation fluid to the turbine.
 18. The endoscopic tool of claim 13, further comprising a connector that connects the endoscopic tool to the instrument channel of the endoscope.
 19. The endoscopic tool of claim 13, wherein the cutting tool comprises a rotor and a casing, the casing having an entry port for supplying the fluid to the turbine and an exit port for removing the fluid from the turbine.
 20. The endoscopic tool of claim 13, further comprising a first controller for selectively supplying the fluid to the cutting tool.
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